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March 27, 1979
SUPERSEDING
Int. Fed. Spec.
00-I-001874A(GL)
May 2, 1978 and
Fed. Spec. 00-I-1874
April 12, 1974

FEDERAL SPECIFICATION

IRONERS, FLATWORK, LARGE ROLL, COMMERCIAL (APRONLESS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers commercial large roll steam heated apronless flatwork ironers.

1.2 Classification. The ironers shall be of the following types, classes, grades and sizes, as applicable, as specified (see 6.2).

Type I - 125 psi (862 kPa) gage minimum operating steam pressure at 345°F (174°C) minimum temperature and condensate trap with valve and by-pass

Type II - Operating steam pressure and temperature, as specified (see 6.2)

Class 1 - With steam inlet pressure and temperature gages
Class 2 - Without inlet pressure and temperature gages

Grade A - Condensate trap with valve and by-pass
Grade B - Condensate trap without valve and by-pass

Size 1 - 655 sheets/hour
Size 2 - 910 sheets/hour
Size 3 - 1080 sheets/hour
Size 4 - 1310 sheets/hour

FSC 3510

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
DDD-S-281	-	Sheet, Bed, Cotton, and Polyester-Cotton

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-L-3153	-	Laundry and Dry Cleaning Machinery and Equipment (For Fixed Installations), Preparation For Delivery Of
MIL-P-43219	-	Paddings and Covers For Flatwork Ironer Roll

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment

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MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-1474 - Noise Limits for Army Material

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

29 CFR 1910 - Occupational Safety and Health Act (OSHA) Standards

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Electrical Manufacturers' Association (NEMA)

ICS - Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York, NY 10017.)

Underwriters Laboratories Inc. (UL) Standards

(Application for copies of applicable standards should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I., NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Boulevard, Tampa, FL 33619.)

National Fire Protection Association (NFPA)

No. 70 - The National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

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American Association of Textile Chemists and Colorists

Test Method 124

Three - Dimensional Durable Press Replicas

(Application for copies of the replicas should be addressed to the American Association of Textile Chemists and Colorists, Research Triangle Park, NC 27709.)

American Welding Society Code (AWS)

Standard Qualification Procedures of the American Welding Society

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

American Society of Mechanical Engineers Code (ASME)

Boiler and Pressure Vessel Code, Section VIII, Pressure Vessels

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The ironers and accessories delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements.

3.2 Codes and standards. The ironers shall comply with the applicable requirements of NEMA ICS, UL, NFPA No. 70, ASME, and OSHA 29 CFR 1910.

3.2.1 Compliance. Prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the ironers and accessory item that he proposes to furnish under this specification meet the applicable requirements of NEMA ICS, UL, NFPA No. 70, ASME, and OSHA.

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3.2.2 NEMA. Acceptable evidence of meeting the requirements of NEMA shall be a certificate of compliance certifying that motor controllers, switches, relays and time delays conform to ICS.

3.2.3 UL. Acceptable evidence of meeting the requirements of UL shall be a certificate of compliance certifying that the wiring, fittings, and conduit conform to the applicable UL standard.

3.2.4 NFPA. Acceptable evidence of meeting the requirements of NFPA shall be a certificate of compliance certifying that ironers and accessories wiring conforms to NFPA No. 70.

3.2.5 ASME. Acceptable evidence of meeting the requirements of ASME shall be a written certificate stating that each chest, chamber, and cylinder has been inspected by an inspector qualified under the provisions of the ASME code for pressure vessels and that the ASME code official stamp has been placed on each chest, chamber, and cylinder by the inspector. The stamp on each chest, chamber, and cylinder shall contain the U or UM symbol, as applicable.

3.2.6 OSHA. Acceptable evidence of meeting the applicable safety and health standards of OSHA shall be a certificate of compliance certifying that the ironers furnished conform to OSHA Standard 29 CFR 1910.264.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial ironer provided that the completed item complies with all provisions of this specification (see 6.5).

3.3.1 Pipe, tube, fittings, traps, and valves. Pipe, tube, fittings, traps, and valves used for air, steam, and condensate lines shall be standard commercial items.

3.3.2 Conduit and electric wire. Electrical conduit shall be rigid electrical metallic tube and flexible metallic conduit. All electric wire shall be stranded or solid copper, heat-resistant grade, thermoplastic insulated.

3.3.3 Motors. Motors shall conform to the requirements of CC-M-1807 and the following:

(a) All motors shall have windings impregnated to resist moisture.

(b) Each motor shall have the proper starting characteristic and ample power with a reasonable factor of safety, commensurate with the application and expected duty cycle to which each motor might be subjected under normal operating conditions.

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(c) Motors shall be equipped with ball bearings, except that motors of 1/4 horsepower (hp) (187 W) or less when used in horizontal applications may have sleeve bearings.

(d) Motor bearings shall be of the permanently lubricated type or shall have adequate and accessible means for lubrication.

(e) Each motor of 1/8 hp (93 W) rating or larger shall be provided with thermal-overload protection incorporated as a part of the motor or installed in the controller.

3.3.4 Motor control circuit. Each motor on ironers and accessory items, as applicable, shall be provided with a controller of the magnetic across-the-line type having a separate pole for each ungrounded conductor. Controllers for direct current motors of more than 1/2 hp (373 W) shall be of the reduced voltage type having one or more steps of armature resistance for smooth acceleration. Pushbutton control stations shall be mounted at points convenient to the operator. Pushbuttons shall be protected or recessed to prevent accidental operation. Timers, limit switches, remote control pushbutton switches, signals, and indicating lights shall be provided to perform the intended functions.

3.3.5 Belts and pulleys. V-belts shall be of the grade commercially known as multiple-drive type. For a fractional horsepower motor where a single belt is used, a V-belt commercially known as a fractional horsepower belt may be supplied. Pulleys for V-belts shall be suitably grooved and shall not be of a die cast type when used in powered drive. Chain type belts shall be of the silent or roller-chain type. Belt and chain tension shall be adjustable.

3.4 Design and construction.

3.4.1 Electrical characteristics. The electrical characteristics shall be as specified (see 6.2).

3.4.2 Wiring. Wiring shall be provided in rigid conduit. Where belt or drive adjustment is required flexible metallic conduit shall be provided. All wiring shall terminate in a connection box with provisions for connections beyond the ironer.

3.4.3 Safety devices. Belts, chains, shafts, pulleys, and gears shall be inclosed or guarded. Guards shall be cast iron or sheet metal. All parts of the guard shall be removable without disassembling pipes or fittings on the ironers. There shall be access openings with covers over parts requiring adjustment or lubrication.

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3.4.4 Equipment base. When specified (see 6.2), the equipment base shall be provided with a leveling device. There shall be provisions for anchoring the base to the floor, and anchor bolts shall be furnished.

3.4.5 Working pressures. Cylinders, chests, and chambers for type I ironers using steam shall be designed for a working pressure of 125 pounds per square inch (psi) (862 kPa) gage and shall be capable of withstanding a hydrostatic pressure specified in the ASME Boiler and Pressure Vessel Code (see 4.4.1). Cylinders, chests, and chambers for type II ironers using steam shall be designed for a working pressure as specified (see 1.2). Class 1 shall be provided with gages installed at the ironer inlet to indicate steam pressure and temperature. Class 2 shall not be provided with gages installed at the ironer inlet to indicate steam pressure and temperature.

3.4.6 Gears and sprockets. Gears and sprockets shall have machined teeth. The use of die-cast gears or sprockets shall not be permitted on any powered drive. Gears shall be of the ferrous or aluminum bronze material and sprockets shall be of ferrous metal. Noise reducing material of phenolic laminate may be used for drive and intermediate pinions. All pinions shall have faces at least as wide as the mating gears.

3.4.7 Bearings and lubrication. Removable, aluminum bronze sleeve, ball, or roller bearings shall be provided for all shafts, rolls, and studs. Bearings shall be replaceable and shall be permanently lubricated unless oil cups or pressure grease fittings are provided. Gears and chains shall be oil or grease lubricated. Seals shall be provided to prevent the lubricant from entering parts of the machine where clothes are handled. Lubrication instructions including location, type of lubricant, and time intervals shall be indented or embossed on a noncorrodible metal plate and permanently affixed in accordance with the manufacturer's practice.

3.4.8 Piping. Integral piping, fittings, pressure reducing valves for air and steam, and drains shall be installed or furnished as complete assemblies suitable for installation with unions or other standard fittings. Type I and type II, grade A machines shall have a condensate trap provided with a valve and by-pass. Type II, grade B machines shall have a condensate trap without a valve and by-pass.

3.4.9 Valves. Inlet steam and air supply connections shall be provided with valves. Steam, air drain, and other operational valves shall be accessible for operation and maintenance.

3.4.10 Spacing and extent of fasteners. The spacing and extent of welds, rivets, bolts, and screws shall insure fastening and prevent bulging of fastened metals.

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3.4.11 Frames. Ferrous-metal frames with legs shall support the rolls, steam chests, and control mechanism at the operating level. The frame shall be connected with rods or structural shapes of ample strength to support the ironing components, and prevent distortion of support members.

3.4.11.1 Rolls. Rolls shall be seamless, ferrous tube, pipe, or spiral welded steel tube with ferrous metal heads or steam heated cylinders and shall be driven through gears to sprockets and roller chains. The rolls shall operate in bearing mounts with springs, or pneumatic or hydraulic cylinders arranged to exert the ironing pressure, or the rolls shall float on the steam chests exerting a minimum pressure of 3/4 psi (5.2 kPa) due to weight of the rolls. The rolls shall be mounted on levers or slides that shall elevate the rolls above the steam chests a minimum of 7/8 inches (22 mm) unless equipped with chests which raise and lower pneumatically. The lifting of the rolls shall be manually controlled. When steam heated cylinders are furnished, the rolls shall be heavy gauge polished steel and shall operate in mounted bearings and shall be driven through gears or sprockets and roller chains. Minimum diameter and length of rolls shall be 19 inches (482 mm) and 120 inches (3050 mm), respectively.

3.4.11.2 Padding and covering. When specified (see 6.2), spring padding or metal mesh padding shall be used on rolls with steam chests. Metal mesh padding shall conform to the requirements of MIL-P-43219. Spring padding shall be equipped with a liner and cover. When flatwork ironer rolls are equipped with metal mesh or spring padding, there shall be a vacuum attachment for exhausting moisture from the padded rolls. Steam heated cylinders shall not require padding.

3.4.11.3 Chests, steam. Steam chests, when furnished, shall be ferrous metal. Chests shall have a smooth machine finish with ground and polished ironing surfaces. For each roll the ironer shall have one steam chest attached to the frame. The chests shall be concave on top to suit the rolls.

3.4.11.4 Shafts. Ferrous metal shafts shall be provided for mounting of pulleys, gears and rolls.

3.4.11.5 Feed. The feed shall consist of ribbons traveling over a power driven roll and a stationary plate member fitted with guiding pins or metal fingers. The ribbons shall be endless, 2-1/2 to 3 inches (64 to 76 mm) wide made of heavy feed-ribbon stock. The ribbons shall be arranged to suit the width of the ironer and to carry the items to be ironed to the first padded roll. There shall be a padded finger roll placed in such a manner to carry goods being ironed to the first padded ironing roll. The finger roll shall rest on the feed ribbons and be powered by the motion of the feed ribbons. Means shall be provided for tightening the ribbons and drive chain if used. The feed device shall be supported by ferrous-metal brackets bolted to the front of the machine.

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3.4.11.6 Feed guard. A metal guard shall be fitted across the entire line of feed, arranged so that a movement of guard in the feeding direction will cause the machine to stop when tested as specified in 4.5.2. The guard shall be located so that the operator's fingers cannot pass between the guard and the ribbon feed. The height of the guard shall be sufficient to prevent the operator or any other person reaching over it and getting his fingers caught between the chest and the first padded roll.

3.4.11.7 Roll guard. When specified (see 6.2), the rolls shall be guarded so that a person cannot reach into the rolls without removing the guard. The guard may be vertical on all sides or a complete cover. If a vertical guard is used, the distance from the floor or working platform to the top of the guard, shall not be less than 6 feet (1.8 meters).

3.4.11.8 Speed indicator. A speed indicator showing the approximate peripheral speed of the rolls in feet per minute (meters per minute) shall be located so as to be readable from the feed side of the ironer.

3.4.11.9 Guides for chest type. Heat-resisting strings or tapes operating over spools or guides shall be provided for guiding the items under the rolls. A string or tape rod shall be fitted at the rear of the machine. Means shall be provided for automatically tightening the strings or tapes. Guides for steam heated cylinder type shall be specially treated ribbons which shall convey the items in a sliding motion against the first heated cylinder and then to subsequent heated cylinders and then to rear of ironer.

3.4.11.10 Motors, controls and operating speeds. The motor shall be started by a push button located at the front of the ironer and stopped by either of two push buttons located one at the front and one at the rear of the ironer. Motors shall be equipped with a two speed drive and a magnetic brake. Ironers shall operate at 25-75 feet (8-23 meters) per minute on low speed and 70-200 feet (21-61 meters) per minute on high speed when tested as specified in 4.4.2. When furnished, the mechanical speed changer controls shall be mounted on a metal bracket on the ironer or on a metal floor stand at operating level at the feed end of the ironer. Means shall be provided for operating the machine at forward speeds for repadding, when ironers have padded rolls. When specified (see 6.2), electrical means shall be provided for operating the drive in the reverse direction. Type II shall be provided with oil pans located underneath the motor and drive assembly.

3.4.11.11 Ventilating canopy. A ventilating canopy conforming to the manufacturer's standard method of exhausting heat from the rolls shall be furnished.

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3.4.11.12 Electromagnetic compatibility. When specified (see 6.2), equipment shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.4).

3.4.11.13 Moisture and fungus resistant treatment. When specified (see 6.2), electrical components of equipment covered by this specification shall be moisture and fungus-resistant treated with material conforming to and applied as required by MIL-T-152.

3.4.11.14 Noise limits. The ironer shall meet the noise limits specified in MIL-STD-1474 and where applicable, shall not exceed 85 decibels A-weighted at the normal operating position when tested as specified in 4.4.5.

3.4.12 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.4.13 Government-furnished property. The sheets, pillowcases, and surgical wrappers required for type II ironer will be furnished by the Government (see 6.2).

3.5 Performance. The ironers shall be operated by electric motors with push button controls.

3.5.1 Ironer, type I. The type I, ironer shall operate on 125 psi (862 kPa) gage minimum steam pressure at a minimum of 345°F (174°C). Ironers shall be capable of ironing sheets, when operated at the speeds and productions shown in table I (see 4.4.3.1).

TABLE I. Ironer (type I)

Size	Speed, ft/min(m/min) up to	Performance sheets/hr
1	90 (27)	655
2	120 (37)	910
3	150 (46)	1080
4	190 (58)	1310

3.5.2 Ironer, type II. The type II ironer shall be capable of meeting the minimum standards, of both quantity and quality, using the minimum specified steam pressure and temperature as measured at the ironer steam inlet when the ironer is operated under ambient conditions of temperature and humidity during working hours in the laundry in which the ironer is installed. The ironer shall be of the size specified and capable of speeds and production as shown in table II. Appearance of the finished flatwork items shall have a minimum average quality rating of 4.0 for sheets and pillowcases and 3.0 for surgical wrappers, and a minimum acceptable moisture removal from 35 $\pm$ 2 percent of bone dry weight before ironing to not more than 2 percent of bone dry weight following ironing when the ironer is tested as specified in 4.4.3.2.

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TABLE II

Size	Speed ft/min (m/min)	Performance (items/lane/hr.)			
		Cotton/polyester blend 65/35 or 50/50		Double thickness Cotton surgical wrappers	
		Sheets 72 X 108 in. (1830 X 2743 mm)	Pillow cases 20-1/4 X 32-1/2 in. (521 X 826 mm)	36 X 36 in. (914 X 914 mm)	24 X 24 in. (610 X 610 mm)
1	90(27)	655	540	485	719
2	120(37)	910	750	675	1000
3	150(46)	1080	890	801	1186
4	190(58)	1310	1079	971	1439

3.6 Finish and color. Unless otherwise specified (see 6.2), each ironer shall be finished in accordance with manufacturer's commercial practice. The color of the emergency stop and normal stop push buttons and feed guard shall be red.

3.7 Marking for identification. Marking for identification of the ironers shall be in accordance with the manufacturer's commercial practice and shall include manufacturer's name, date of manufacture, model number and electrical characteristics.

3.8 Workmanship. The completely fabricated item shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, miscasts, deterioration or malformations. There shall be no sharp edges, burrs or slivers.

3.8.1 Application of finish. The finish applied to the end item shall be continuous, smooth, adherent, have no foreign matter imbedded and contain no sags, runs, drips, creeps, laps, bubbles or streaks. No rust or tool marks shall show through the coating.

3.8.2 Welders and welding.

3.8.2.1 Welders. The welders performing the welding on subassemblies, and on the end item shall be certified welders and have passed the qualification test as prescribed by the Standard Qualification Procedures of the American Welding Society or Welding Qualifications of the American Society of Mechanical Engineers Code. The contractor will furnish the Government with a list of names of his welders who are certified and have passed the test of either referenced codes and he shall certify that the welding of this equipment was performed by the welders listed.

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3.8.2.2 Welding and brazing. All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fractures or incomplete fusion. All scale and flux deposits shall be removed from finished welds.

3.8.3 Soldering. Soldering shall be complete and adherent with all flux and residue removed and shall contain no pinholes.

3.8.4 Bolts, nuts, screws, studs, and other threaded fasteners. Threaded fasteners shall not be broken, stripped, fractured or loose.

3.8.5 Rivets. Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface of the members and shall not be loose.

3.8.6 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.2.2 End item inspection. The inspection lot shall be all ironers of the specified size, offered for inspection at one time.

4.2.2.1 Visual inspection. The ironer shall be examined for the defects in table III. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE III. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished where required	X	
	Finish and color not as specified	X	
	Scale on corrosion-resisting metal or plated metal		X
	Emergency stop and normal stop push buttons not red color or feed guard not red color	X	
	Not continuous, smooth, adherent, foreign matter imbedded and sags, runs, drips, creeps, laps, bubbles or streaks		X
	Rust or tool marks		X
Construction	Not type, class, grade, size, specified	X	
	Any part missing or not as specified	X	
	Split, punctured, torn, dented, creased, deteriorated or malformed	X	
	Sharp burr, sliver, or sharp edge	X	
	Push button not protected or recessed	X	
	Exposed belts, chains, shafts, pulley or gears	X	
	Any functioning components that requires abnormal force to operate	X	
	Bearings not replaceable	X	
	Guards not removable without disassembling pipes or fitting	X	
	Base not in full contact with a level floor	X	
	No provision for anchoring base to floor		X
	Anchor bolts not provided		X
	Component not connected or joined as specified	X	

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TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction (cont'd)	Pinion not as wide as meshing gear		X
	Seals not provided to prevent grease from entering parts of machine where flatwork is handled	X	
	No means to adjust belt and chain tension	X	
	Not apronless type	X	
	Control mechanism not at operating level	X	
	No leveling devices	X	
	Rolls on cylinders not seamless or spiral welded		X
	No springs or pneumatic or hydraulic cylinders to exert ironing pressure	X	
	Not a steam chest for each padded roll, when applicable	X	
	Guard across line of feed missing or allows operator's fingers to pass through	X	
	Roll sections cannot be added	X	
Welding and brazing (where applicable)	Missing, incomplete, burn holes, fractured or not fused	X	
	Slag inclusion, undercut, not smooth and uniform, or deposit of scale and flux not removed	X	
Soldering (where applicable)	Missing, incomplete or not adherent	X	
	Not clean (flux or residue not removed) or contains pinholes		X
Bolts, nuts, screws, studs and other threaded fasteners	Missing, broken, stripped or fractured	X	
	Threaded fasteners loose		X
Rivets (when applicable)	Missing, broken, not peened or not peened sufficiently	X	
	Rivets loose		X

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TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Electrical apparatus wiring	Wiring cut, abraded, excessive insulation stripped, not properly joined or loose at terminals	X	
	Wiring not terminating in junction box	X	
	Not inclosed in specified conduit where required	X	
	Adequate slack not provided for wiring to relieve strain		X
Piping, tubing, fittings and valves	Improperly connected or installed; will not provide a seal or threads stripped	X	
Lubrication	Not lubricated where required		X
	Excessive lubrication		X
	Fitting not accessible for lubrication		X
Marking for identification	Missing; incomplete, illegible, not specified type or size; or not in proper location		X

4.2.2.2 Dimensional examination. The ironer and accessory item shall be examined for compliance with dimensions specified. Any dimension not within this specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. The first produced ironer shall be tested as specified in 4.4.1, 4.4.3, 4.4.4 (when applicable) and 4.4.5. Each sample shall be tested as specified in 4.5.2. Failure to pass a test shall be cause for rejection of the inspection lot. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed as defects per hundred units.

4.3 Packaging inspection. Preservation, packing and marking shall be inspected in accordance with the quality assurance provisions of MIL-L-3153.

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4.4 Tests.

4.4.1 Hydrostatic pressure. Assemblies of cylinders and chambers that in operation contain steam shall be subjected to a hydrostatic pressure test determined in accordance with the provisions of the ASME Code for Unfired Pressure Vessels based on a design pressure of not less than 125 psi (862 kPa) gage for type I ironers and as specified (see 6.2) for type II. The pressure shall be maintained a sufficient length of time to determine compliance with 3.4.5. Any leaks or rupture shall constitute failure of this test.

4.4.2 Operational test. The ironer shall be operationally tested for compliance with the requirements of 3.4, 3.4.11.6, and 3.4.11.10.

4.4.3 Functional test.

4.4.3.1 Ironer, type I test. Connect the ironer to electrical power and to steam line at 125 psi (862 kPa) gage minimum steam pressure, and minimum temperature of 345°F (174°C) and operate at a test speed of not less than speeds specified in table I, for the specified size ironer. Sheets conforming to type III or IV, size 7 of DDD-S-281, and containing 35 ± 2 percent of their bone dry weight in water shall be ironed in a time period not to exceed 1 hour. The sheets per hour to be ironed shall be as specified in table I for the size ironer specified. Examine the sheets to determine that moisture retention does not exceed 2 percent of their bone dry weight in conformance with the requirements specified in 3.5.1. Any nonconformance shall be cause for rejection.

4.4.3.2 Ironer, type II test. The ironer shall be connected to utilities providing specified electrical power, steam pressure, and steam temperature. Spreader-feeders and folders shall not be used. Items listed in table II shall be used in the test. Testing shall be performed under ambient temperature and humidity in the laundry during working hours. Items used for testing shall be items that have had prior use. Items shall be weighed after being completely dried in a drier conditioner to determine bone dry weight. Items shall be washed and extracted or washed, extracted and conditioned and then weighed to achieve and determine 35 ± 2 percent moisture content. Conditioned weight minus bone dry weight divided by bone dry weight times 100 equals percentage of moisture retention. Items are to be processed through the flatwork ironer in test lots to be run 60 minutes. A stop watch will be used to determine the time of the tests. The number of items used in the test will be determined by actual count. As items are processed through the flatwork ironer, a random sample of the item will be selected from each test lot and rated for appearance. This rating shall be in accordance with the Evaluation Section of the AATCC Test

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Method 124 except the rating shall be performed by three VA designated observers in lieu of three trained observers as specified in Test Method 124. Specimens shall be pinned to the rating board for viewing in such a manner that they are allowed to hang freely without stretching. The area of the sample sheets to be viewed shall be not less than 15 x 15 inches (381 x 381 mm) and shall be no closer than 6 inches (152 mm) from any edge. The area for all other items shall be 15 x 15 inches (381 x 381 mm) centered. The resulant thirty observations, three judgements on each test specimen, shall be averaged and the result reported to the nearest tenth of a rating point. Appearance of the finished flatwork items with an average rating between 0.0 and 3.9 inclusive for sheets and pillowcases and 0.0 and 2.9 inclusive for surgical wrappers will be considered unacceptable quality. At the conclusion of each test run the processed items shall be weighed to determine that moisture retention does not exceed 2 percent of their bone dry weight. Failure of the flatwork ironer to produce the minimum in quality or quantity, or to reduce water retention to within 2 percent of bone dry shall be cause for rejection.

4.4.4 Electromagnetic compatitility. When electromagnetic compatibility is required, the first unit of production shall be tested by the contractor in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The supplier shall furnish the contracting officer a written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Compatibility Test Report and the requirements meet MIL-STD-461 (see 3.4.11.12).

4.4.5 Noise test. The ironer shall be tested for conformance with the noise limits of 3.4.11.14 in accordance with MIL-STD-1474.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with levels A, B, and C requirements of MIL-L-3153, as specified (see 6.2).

6. NOTES

6.1 Intended use. The ironer covered by this specification is intended for use in fixed installations.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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(a) Title, number, and date of this specification.
(b) Type, class, grade and size, as applicable, as required (see 1.2).
(c) Electrical characteristics required (voltage, frequency, and number of phases) (see 3.4.1).
(d) When a leveling device is required (see 3.4.4).
(e) Whether spring padding or metal mesh padding is required (see 3.4.11.2).
(f) When rolls shall be guarded (see 3.4.11.7).
(g) When electrical reversing of drive is required (see 3.4.11.10).
(h) When electromagnetic compatibility is required (see 3.4.11.12).
(i) When moisture and fungus treatment is required (see 3.4.11.13).
(j) Contract data requirements (see 3.4.12 and 6.2.1).
(k) Source of Government furnished property (see 3.4.13 and 6.3).
(l) Finish when other than specified (see 3.6).
(m) Levels of preservation, packing, and marking required (see 5.1).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity (see 3.4.12).

6.3 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.4.13.

6.4 Successful commercial application. Invitation for bids should specify that no item of equipment shall be acceptable unless the manufacturer has had equipment of approximately the same type and design as that specified operating successfully in a commercial or institutional laundry or laundries for at least one year. Equipment installed for test purposes in a manufacturer's plant or laboratory shall not come within the category of successful commercial application.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.6 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

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Custodians:

Army - GL
Air Force - 99

Review activity:

Army - MD

User activity:

Navy - MC

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA - FSS
VA - DMS

Project No. 3510-0272

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